

Type: MCQ

Q1. What is data visualization? (1)

1. Representing data in the form of tables
2. **Presenting data in graphical or visual formats
3. Summarizing data using descriptive statistics
4. Analyzing data through machine learning algorithms

Q2. Which of the following is NOT a type of data visualization? (1)

1. Line chart
2. Histogram
3. **SQL query
4. Scatter plot

Q3. What is the purpose of data visualization? (1)

1. To make data more complex
2. **To simplify complex data and make it understandable
3. To hide patterns in the data
4. To limit access to data

Q4. Which of the following is NOT a benefit of data visualization? (1)

1. Improved decision-making
2. Enhanced storytelling
3. **Increased complexity of data
4. Identification of trends and patterns

Q5. What does the mean represent in a dataset? (1)

1. The most frequently occurring value
2. The middle value when the data is sorted
3. **The average value of all data points
4. The difference between the highest and lowest values

Q6. The mode of a dataset is: (1)

1. The middle value
2. **The value that occurs most frequently
3. The average value
4. The difference between the maximum and minimum values

Q7. What does the median represent in a dataset? (1)

1. The average value
2. **The middle value when the data is sorted
3. The most frequently occurring value
4. The range of values

Q8. Which measure of dispersion indicates the spread of values around the mean? (1)

1. Mode
2. Median
3. **Standard deviation
4. Range

Q9. What does the range of a dataset represent? (1)

1. The difference between the mean and median
2. The average value
3. **The spread between the largest and smallest values
4. The sum of all values in the dataset

Q10. What is the purpose of Tableau Prep? (1)

1. Creating visualizations
2. Connecting to data sources
3. **Data preparation and cleaning
4. Publishing dashboards

Q11. Which of the following is NOT a type of chart that Tableau can create? (1)

1. Line chart
2. Bar chart
3. **3D modeling
4. Pie chart

Q12. What is Tableau Public? (1)

1. A paid version of Tableau
2. **A free version of Tableau for public sharing
3. A data storage platform
4. A programming language

Q13. Which file format does Tableau use to save workbooks? (1)

1. .xls
2. .txt
3. **.twb
4. .csv

Q14. In Tableau, what does the term "Dashboard" refer to? (1)

1. A single data visualization
2. **A collection of multiple visualizations and sheets
3. A data connection
4. A data source

Q15. What is a dimension in Tableau? (1)

1. **A categorical field used for grouping and organizing data
2. A numerical field used for calculations
3. A filter applied to data visualizations
4. A color palette for charts

Q16. Which of the following is a measure in Tableau? (1)

1. Category
2. **Sales revenue
3. Region
4. Date

Q17. How are dimensions represented in Tableau visualizations? (1)

1. With numerical values
2. **As categorical labels or headers
3. As calculated fields
4. With color gradients

Q18. Which type of data is typically used as a measure in Tableau? (1)

1. Date
2. Category
3. **Numeric values
4. Text values

Q19. What is the primary purpose of dimensions in Tableau? (1)

1. Performing calculations
2. Grouping and categorizing data
3. Applying filters
4. Defining color schemes

Q20. What is the purpose of a filter in Tableau? (1)

1. To create visualizations
2. **To control which data is displayed in a visualization
3. To calculate measures
4. To format text labels

Q21. What type of data includes timestamps indicating when events occurred? (1)

1. Machine data

2. Structured data
3. **Time-stamped data
4. Real-time data

Q22. Which type of data is generated by sensors, devices, or machines? (1)

1. Time-stamped data
2. **Machine data
3. Spatiotemporal data
4. Open data

Q23. What is structured data? (1)

1. Data with timestamps
2. **Data organized into rows and columns
3. Data from sensors
4. Data in real-time

Q24. Which type of data lacks a predefined data model and is not organized in a structured manner?
(1)

1. Structured data
2. **Unstructured data
3. Semi-structured data
4. Real-time data

Q25. What characterizes semi-structured data? (1)

1. Data in real-time
2. Data with timestamps
3. **Data with some organization but not rigidly structured
4. Data from sensors

Q26. What does spatiotemporal data refer to? (1)

1. Data with timestamps

2. **Data that includes both spatial and temporal components
3. Data from machines
4. Real-time data

Q27. What is open data? (1)

1. Data with timestamps
2. **Data that is freely available for anyone to use, reuse, and redistribute
3. Data from sensors
4. Real-time data

Q28. Which type of data is generated and updated in real-time? (1)

1. Time-stamped data
2. Open data
3. **Real-time data
4. Machine data

Q29. What is operational data? (1)

1. Data with timestamps
2. **Data used in the day-to-day operations of an organization
3. Data from machines
4. Real-time data

Q30. Which type of data is often stored in databases and can be easily queried and analysed? (1)

1. Semi-structured data
2. Open data
3. **Structured data
4. Machine data

Q31. Which type of data is commonly found in documents, emails, social media posts, and multimedia files? (1)

1. Time-stamped data

2. Structured data
3. Semi-structured data
4. **Unstructured data

Q32. What is the primary challenge associated with unstructured data? (1)

1. Difficulty in collecting data
2. Lack of real-time updates
3. **Complexity in organizing and analyzing the data
4. Limited data volume

Q33. What is the first step in the data analytics and visualization lifecycle? (1)

1. Data collection
2. Data cleaning
3. **Defining the problem and objectives
4. Data visualization

Q34. Which phase involves exploring and understanding the data to identify patterns and trends?
(1)

1. Data cleaning
2. Data modeling
3. **Data exploration and analysis
4. Data visualization

Q35. What is the purpose of data cleaning? (1)

1. Defining the problem
2. Data exploration
3. **Removing inconsistencies and errors from the data
4. Data visualization

Q36. Which phase involves transforming raw data into a format suitable for analysis and visualization? (1)

1. Data exploration
2. Data cleaning
3. **Data preprocessing
4. Data modeling

Q37. What is data modeling in the data analytics lifecycle? (1)

1. Defining the problem
2. Data exploration
3. Data cleaning
4. **Creating statistical or machine learning models

Q38. Which phase involves creating visual representations of data to communicate insights? (1)

1. Data preprocessing
2. Data modeling
3. **Data visualization
4. Data exploration

Q39. Why is defining the problem and objectives important in data analytics? (1)

1. To collect more data
2. To explore data visually
3. **To ensure focus and relevance in the analysis
4. To clean data

Q40. How does data visualization contribute to data analytics? (1)

1. By collecting more data
2. By cleaning data
3. **By making insights understandable and actionable
4. By defining the problem

Q41. Which type of chart is best suited for comparing proportions or percentages of a whole? (1)

1. Line chart

2. Scatter plot
3. Bar chart
4. **Pie chart

Q42. What type of chart is used to show the distribution of numerical data across categories? (1)

1. Pie chart
2. **Bar chart
3. Line chart
4. Scatter plot

Q43. Which chart is suitable for visualizing trends or changes over time? (1)

1. Bar chart
2. **Line chart
3. Pie chart
4. Histogram

Q44. What is the purpose of a scatter plot? (1)

1. Comparing proportions
2. **Showing the relationship between two numerical variables
3. Displaying distributions
4. Visualizing trends over time

Q45. What is the primary use of a histogram? (1)

1. Showing trends over time
2. **Displaying the frequency distribution of numerical data
3. Comparing proportions
4. Visualizing hierarchical data

Q46. Which chart type is suitable for comparing individual data points across categories? (1)

1. Line chart
2. Scatter plot

3. **Column chart
4. Pie chart

Q47. What is a slicer in Power BI? (1)

1. A tool for writing DAX expressions
2. **A filter control used to interactively filter data
3. A data storage component
4. A visualization type

Q48. Which tool in Power BI allows users to create custom calculations and measures? (1)

1. Charts
2. Tiles
3. **Data Analysis Expressions
4. Filters

Q49. How can you customize the layout and design of a Power BI dashboard? (1)

1. By changing the color scheme
2. By adding text boxes
3. **By using the Power BI desktop layout view
4. By resizing visualizations

Q50. Which tool in Power BI allows users to create relationships between different data tables?
(1)

1. Charts
2. Tiles
3. **Power BI Desktop
4. Filters

Q51. What is the difference between a Power BI dashboard and a Power BI report? (1)

1. Dashboards allow data exploration, while reports display static data.
2. **Dashboards provide a high-level overview, while reports offer detailed analysis.

3. Dashboards are interactive, while reports are static.
4. Dashboards are designed for end-users, while reports are for data analysts.

Q52. Which Excel feature allows you to organize data into rows and columns? (1)

1. Charts
2. **Spreadsheets
3. Formulas
4. Conditional formatting

Q53. What function is used to sum a range of cells in Excel? (1)

1. CONCATENATE
2. **SUM
3. VLOOKUP
4. IF

Q54. What does the function COUNTA do in Excel? (1)

1. Counts the number of cells that meet a specific condition
2. Counts the number of empty cells in a range
3. **Counts the number of non-empty cells in a range
4. Counts the number of errors in a range

Q55. What is the purpose of conditional formatting in Excel? (1)

1. Sorting data
2. **Highlighting data based on specific criteria
3. Creating charts
4. Calculating totals

Q56. How can you freeze panes in Excel? (1)

1. By hiding rows and columns
2. **By locking rows or columns to keep them visible while scrolling
3. By merging cells

4. By using conditional formatting

Q57. Which Excel feature is used to create visually appealing representations of data? (1)

1. Pivot tables
2. Formulas
3. **Charts
4. Conditional formatting

Q58. What is an absolute reference cell in Excel? (1)

1. A cell that cannot be edited
2. A cell with a fixed value
3. **A cell reference that does not change when copied or moved
4. A cell with conditional formatting

Q59. What does a relative reference cell do in Excel? (1)

1. It references a cell in another worksheet
2. **It adjusts its reference when copied or moved to a new location
3. It highlights cells based on specific criteria
4. It calculates the average of a range of cells

Q60. Which type of cell reference is recommended when you want to keep a cell reference constant in a formula? (1)

1. Relative reference
2. Fixed reference
3. **Absolute reference
4. Dynamic reference

Q61. How can you create an absolute reference to cell A1 in an Excel formula? (1)

1. By typing =A1
2. By using the formula =SUM(A1)
3. **By using the formula =\$A\$1

4. By using the formula =A\$1

Q62. Which of the following is/are the examples of structured data? (1)

1. Videos
2. **Employee's name, employee's id, employee's age
3. Audio files
4. All of the mentioned above

Q63. What is the most effective method of preparing your data for Power BI? (1)

1. **User of a star schema
2. Load all tables
3. Include multiple objects
4. None of the above

Q64. What is the use of data cleaning? (1)

1. To remove the noisy data
2. Correct the inconsistencies in data
3. Transformations to correct the wrong data
4. **All of the above

Q65. To remove noisy and inconsistent data _____ is needed. (1)

1. Data Transformation
2. Data Integration
3. **Data Cleaning
4. Data Reduction